

East West University

Department of Mathematical and Physical Sciences

Course Title: Coordinate Geometry and Vector Analysis, Course code: MAT 104

Semester: Fall 2025, Section: 14

Quiz Test-I, Total Marks: 10 (3+3+4), Duration: 30 min.

Answer all the questions

1. Transform the equation $7x^2 + 8xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving xy .
2. Prove that the equation $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represents two straight lines. Find their point of intersection and the angles between them.
3. Define direction ratios.
Find the direction cosine of the line that is perpendicular to the lines with direction cosines proportional to $(1, -2, -2)$, $(9, -2, 4)$.

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Quiz Test-II, Total Marks: 10 (5+5), Duration: 30 min.
Answer all the questions

1. Find the asymptotes of the hyperbola $x^2 + 2xy - y^2 + 2x + 4y = 0$ and the equation of the conjugate hyperbola.
Write down the equation of a tangent to the Parabola.
2. What is the length of the chord intercepted by the circle $x^2 + y^2 = p^2$ on the line $y = mx + c$.
Find the pole of the lines $2x - y + 10 = 0$ w.r.t the circle $x^2 + y^2 - 7x - 5y - 1 = 0$.

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Answer all the questions

1. Prove that $(\bar{A} \times \bar{B}) \times (\bar{C} \times \bar{D}) = \bar{B}(\bar{C} \cdot \bar{D} \times \bar{A}) - \bar{A}(\bar{C} \cdot \bar{D} \times \bar{B})$. 2

Write down the area of the parallelogram.

2. If $\bar{F} = (2y + 3)\hat{i} + xz\hat{j} + (yz - x)\hat{k}$, evaluate $\int \bar{F} \cdot d\bar{r}$ along the straight lines from $(0,0,0)$ to $(0,0,1)$, then to $(0,1,1)$ and then to $(2,1,1)$.

Define surface integral.

3. In what direction from the point $(1,3,-4)$ is the directional derivative of $\varphi = 2xz^2 - 3xy - 4x - 7$ a maximum? What is the magnitude of this maximum?



EAST WEST UNIVERSITY

Department of Mathematics and Physical Sciences (MPS)

MIDTERM EXAMINATION (SEMESTER: SUMMER 2025)

COURSE CODE: MAT 102-19 TITLE: DIFFERENTIAL EQUATIONS AND SPECIAL FUNCTIONS

TIME: 80 MINUTES

MARKS: 30

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∴ There are SEVEN questions. Answer all questions. Figures in the right-hand margin indicate full marks.

Solve the differential equation: $\frac{dy}{dx} = (2x + 3y - 5)^2$.

[04]

State whether the following differential equation is homogeneous in x and y or not and

[04]

hence using appropriate transformation solve it: $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$.

[03]

Solve the differential equation: $2(3xy^2 + 2x^3)dx + 3(2x^2y + y^3)dy = 0$.

Test whether the following differential equation is exact or not. If not exact, find its integrating factor and hence solve it: $[y^2(x+1) + y]dx + (2xy + 1)dy = 0$.

[04]

Solve the first order linear differential equation: $(1 + x^2)\frac{dy}{dx} + 2xy = 4x^2$.

[04]

Write down the general form of Bernoulli's differential equation. Find the general

[04]

solution of the differential equation: $\frac{dy}{dx} + \frac{1}{x} \sin 2y = x^3 \cos^2 y$.

Radioactive substances decay at a rate that is proportional to the amount present. The half-life of a substance is the time required for a given amount to be reduced by one-half. The half-life of cesium-137 is 30 years. Suppose we have 100 mg sample.

[04]

- Find the mass that remains after t years.
- How much of the sample remains after 100 years?
- After how long will only 1 mg remain?

Best of Luck